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Original research

Using collaborative learning to improve diabetes care and outcomes: The VIDA project

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ABSTRACT

The prevalence of diabetes in Mexico among those 20–64 years of age has increased from 7.2% in 1993 to 10.7% in 2000. National population-based surveys in Mexico demonstrated that 50% of the total population with diabetes had blood glucose levels of 200 mg/dl or higher. Thus, diabetes care has become one of the most important public health challenges in this country. The aim of the study was to improve the quality of diabetes care in primary health care centers using the chronic care model and the breakthrough series (BTS) collaborative methodology.

Methods: Ten public health centers in the cities of Xalapa and Veracruz were randomly selected to participate in the project. Five of the health centers were randomly assigned to receive the intervention (intervention group) and the other five followed usual care (usual care group). The intervention was evaluated by A1c test before and after the intervention in both groups of patients. Patients were followed for 18 months from November 2002 to May 2004. Results were adjusted for the clustering of patients within practices and baseline measure.

Results: The proportion of people with good glycemic control (A1c < 7%) among those in the intervention group increased from 28% before the intervention to 39% after the intervention. The proportion of patients achieving three or more quality improvement goals

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increased from 16.6% to 69.7% ($p < 0.001$) among the intervention group while the usual care group experienced a non-significant decrease from 12.4% to 5.9% ($p = 0.118$). The focus on the primary care team and the participation of people with diabetes were strategic elements incorporated into the methodology, expected to ensure sustainability of continued improvement of health outcomes. The intervention introduced modifications to solve problems identified by health teams in their practice and improved process and outcome measures of quality diabetes care. Most of the actions were directed at four components of the chronic care model: self-management support, decision support, delivery system design, and clinical information systems.

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1. Introduction

The prevalence of diabetes in Mexico among those 20–64 years of age has increased from 7.2% in 1993 [1] to 10.7% in 2000 [2]. The State of Veracruz presented the highest prevalence of diabetes in Mexico, 16.1% in 2000. In 2000, a study by the Pan American Health Organization (PAHO) estimated that diabetes represented a total expenditure of US$ 15 billion in Mexico; most of which was due to indirect cost caused by premature mortality, and disability [3]. A recent study conducted in Mexico [4] estimated that between 2004 and 2006, the number of cases of diabetes in clinics and hospitals increased from 844,211 to 951,417 (12.7%). This represents an increase in the costs for diabetes care from US$ 56,461,948 (2004), to US$ 63,200,887 (2006), a 14% increase in just 2 years.

National population-based surveys in Mexico indicated that 50% of the total population with diabetes had poor metabolic control (as measured by blood glucose level of 200 mg/dl or higher) [1,2]. Although glycosylated hemoglobin (A1c) is the most accurate parameter to measure metabolic control, the available nationally representative data were based on fasting blood glucose (FBG). Recent priority-setting literature supports the use of FBG when HbA1c is unaffordable [5]; at the time of the surveys A1c was not covered by the public health system in Mexico. As a result of the increased prevalence and poor control of diabetes in Mexico, diabetes mortality has shown a sustained increase during the last decades. While diabetes ranked as the third cause of death in the general population in 1997 with a rate of 38.0 deaths per 100,000 population, it climbed to the first cause of death in 2000, accounting for 46,614 deaths and a reported mortality rate of 46.8 per 100,000 population [6,7].

To evaluate a more integrated approach to diabetes care we conducted a pilot project in the State of Veracruz. The aim was to improve the quality of diabetes care in primary health care centers using the chronic care model proposed by Wagner et al. [8,9] and the breakthrough series (BTS) collaborative methodology developed by the Institute for Healthcare Improvement (IHI) [9].

2. Methods

The project was implemented through the Secretariat of Health of the State of Veracruz, Mexico (SSVER) that provides care for 3,583,931 people which represents approximately half of the total population of the state of Veracruz (as of 2005). SSVER's beneficiaries are mostly people of low resources that do not receive medical benefits from the government social security system or health insurance through private employers.

Ten of the 23 SSVER health centers in the cities of Xalapa and Veracruz in the State of Veracruz were randomly selected to participate in the project. All the 10 centers implemented a clinical information system and all patients with diabetes were offered two glycosylated hemoglobin (A1c) tests and a lipid profile test at baseline and the end of project. Five of the health centers were randomly selected to receive the intervention (the intervention group) and the other five monitored their patients receiving usual care (usual care group). All patients participating in the project gave their written informed consent and the protocol was approved by the state ethical committee. The clinical information system was based on the QUALIDIAB [10] program of the Declaration of the Americas on Diabetes (DOTA). QUALIDIAB had been implemented with success before in several countries of Latin America (Argentina, Brazil, Chile, Colombia, Paraguay, and Uruguay). The clinical information system implemented in Veracruz consisted of paper-based records and a computer database. This database facilitated the preparation and dissemination of data abstracts to health providers to monitor quality of care. In addition to the clinical information system and biochemical tests, all centers provided patients the opportunity to participate in peer support groups.

A total of 43 primary care teams participated in the project; teams were made up of physicians, nurses and, in some centers, other professionals such as nutritionists, and psychologists. The study's outcomes were monitored through the clinical information system that consisted of the periodical review of the clinical charts of the 307 patients involved in the study (196 from the intervention group and 111 from the usual care group).

Participating health providers from the intervention center teams completed the Assessment of Chronic Illness Care (ACIC) questionnaire adapted for diabetes, before and after the intervention to help in the selection of priorities for interventions [9]. The ACIC is a self-evaluation tool through which health providers can identify areas for improvement in their care delivery; it is especially useful to conduct before a team begins a collaborative quality improvement cycle. The adapted version of the ACIC questionnaire in Spanish can be obtained from the authors.

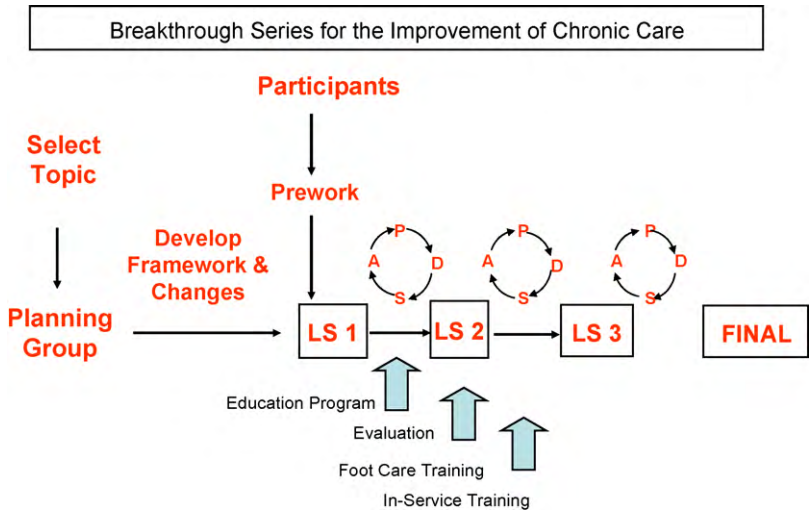

Fig. 1 – The VIDA collaborative diagram.

3. Core intervention

The core intervention consisted of three learning sessions (LS1, LS2 and LS3) using the BTS methodology (Fig. 1). The three learning sessions were organized within a period of 18 months (from November 2002 to May 2004) and consisted of the implementation of three fundamental strategies to improve quality of diabetes care: a structured patient diabetes education program (LS1–LS2), training in foot care (LS2) and in-service training for primary care personnel in diabetes management (LS2–LS3) in all the intervention centers. During the three learning sessions the most important problems in the medical care delivery were analyzed using the chronic care model from Wagner et al. [8] as a framework. Through this model, the primary health care teams, which included primary health care personnel as well as personnel from the local hospital, were trained to adopt a quality improvement methodology that allowed them to identify specific problems within their practice and develop solutions.

In each of the three learning sessions, the teams selected specific objectives for improvement cycles. The objectives were based on those problems found in the practice of each health center. These cycles followed a standard "Plan, Do, Study, Act" (P-D-S-A) of quality improvement practice, adapted to health care delivery by IHI and known as the BTS methodology.

A selected group of patients with diabetes in the centers implementing the intervention participated in the learning sessions. Patients were selected by the medical teams based on their capacity to communicate, their advanced knowledge of diabetes, and their willingness to collaborate. People with diabetes participated in two ways: one, along with members of the health center and two, along with other people with diabetes. In one of the learning sessions, participants with diabetes were requested to perform a qualitative assessment of the support groups so that real-time adjustments could be made where needed reflecting participant feedback.

4. Diabetes education

The educational program was based on the Diabetes Education Program of Latin America (PENID-LA) model [11] developed by the Center of Experimental and Applied Endocrinology (CENEXA) of La Plata, Argentina, a PAHO/WHO Collaborating Center in the area of diabetes. The program materials consist of a flipchart that is used to provide patient education, as well as a patient manual entitled "How to Treat My Diabetes" [12]. The course included four units focused on general knowledge of diabetes (1), nutritional information and management (2), foot care and regular physical activity (3) and diabetes care and sick days (4).

5. Support from specialists

As a part of the intervention, the current referral system in SSVER was modified by an innovative plan that brought the specialists to the health centers. In this plan, the core health team participated in patient consultations together with specialists. These cross-consultations were designed to meet patient management and teaching objectives. Additionally, a group of national and international professionals provided technical advice throughout the implementation of the project and participated in learning sessions.

6. Case management

An advisor provided support for case management by visiting the intervention clinics. In-service training was provided while seeing cases in the clinics. Health providers (physicians and nurses) selected cases that were not achieving goals to be seen by the case management advisor. During the case management sessions, patients and providers set management goals that were reasonable to achieve. Treatment changes were introduced following the Mexican guidelines for the management of diabetes.

In addition to these four components, some of the centers focused on specific activities such as enhanced psychological support, increased physical activity, or the improvement of the patient–provider relationship.

Individual patients discussed goals for care with the health team. Care was oriented toward achieving the following general care targets:

- Metabolic control (A1c < 7%).
- Cholesterol < 200 mg/dl.
- BP < 140/90.
- Foot examination conducted.
- Eye examination conducted.

In addition to the general care targets listed above, each patient may have defined additional targets with his/her health team related to other aspects of diabetes management. Quality improvement was measured by changes in the proportion of patients achieving the general care goals.

7. Analysis

Descriptive results are presented as percents of the total for categorical variables and means and standard deviation (SD) for continuous variables. All measures are broken down in two groups (intervention and usual care groups), before and after the intervention. The Chi-square test was used to demonstrate statistical difference before and after the intervention in both the intervention and the usual care groups. A probability of less than 0.05 was selected as statistically significant. The McNemar test and the two-tailed paired *t*-test were used to demonstrate differences between the groups in categorical and continuous variables, respectively.

We fitted a multilevel logistic regression model with random intercept of the baseline values of each of the six outcomes measured. All models were adjusted for the clustering of patients within practice, age and gender as well as the baseline measure. All analyses were conducted using SPSS15 (SPSS Inc. 2007) and Stata10 (Stata Corporation, 2007).

8. Results

Table 1 shows characteristics of patients in the intervention and usual care groups. Overall 196 patients were evaluated in the centers where the intervention was carried out and 111 patients were evaluated in the centers that provided usual care. The distribution of patients by age, and sex, as well as by type of diabetes (type 1 or type 2) and smoking status, was

Table 1 – Proportion of participants, the intervention group and usual care, according to age and sex.

	Intervention = 196			Usual care = 111		
	Men	Women	Total	Men	Women	Total
Age[a]						
<40	9.1	5.6	6.1	5.9	4.3	4.5
40–59	63.6	52.1	54.1	58.8	56.4	56.8
60+	27.3	42.3	39.8	35.3	39.4	38.7
Within groups *p*-values	0.25			0.92		
Between groups *p*-value	0.80					
Smoking						
Yes	37.1	4.9	10.6	22.2	2.1	5.2
No	62.9	95.1	89.4	77.8	97.9	94.8
Within groups *p*-values	<0.01			<0.01		
Between groups *p*-value	0.36					
Type of diabetes[a]						
Type 1	6.1	1.8	2.6	0.0	1.1	0.9
Type 2	93.9	98.2	97.4	100.0	98.9	99.1
Within groups *p*-values	0.20			0.85		
Between groups *p*-value	0.29					
Number of consultations attended[b]						
0–4	12.5	3.7	5.2	13.3	11.5	11.8
5–9	40.6	40.4	40.4	66.7	40.2	44.1
10+	46.9	55.9	54.4	20.0	48.3	44.1
Within groups *p*-values	0.11			0.11		
Between groups *p*-value	0.06					
Participated support groups[c]						
Yes	87.9	79.8	81.1	11.8	36.2	32.4
No	12.1	20.2	18.9	88.2	63.8	67.6
Within groups *p*-values	0.20			0.04		
Between groups *p*-value	<0.01					

[a] At the beginning of the intervention.
[b] During the year of the intervention.
[c] At the end of the intervention.

Table 2 – ACIC evaluation by area of the chronic care model at the beginning and end of the collaborative intervention.

	Beginning	End	Change	SD	P
Organization of care	4.7	8.6	+3.9	1.4	<0.01
Community linkages	4.3	6.1	+1.7	0.7	0.01
Self-management support	6.2	9.9	+3.7	1.1	<0.01
Delivery system design	5.3	9.4	+4.1	1.4	<0.01
Decision support	7.2	9.6	+2.3	0.8	<0.01
Clinical information system	6.2	8.4	+2.2	2.2	0.08
Total	5.7	8.7	+3.0	0.6	<0.01

similar in the intervention group and the usual care group. Therefore, the random selection of centers was successful. Men were more likely to be classified as smokers than women among both the intervention and usual care groups; 54.4% of the intervention group and 44.1% of the usual care attended consultation 10 or more times during the year of the intervention. The proportion of patients that participated in the support groups was greater in the intervention group (81.1%) than in the usual care group (32.4%) ($p < 0.01$).

Table 2 reports the results of the ACIC questionnaire before and after the intervention. The ACIC questionnaire was completed by 120 and 149 participants at the first and third learning sessions, respectively, representing 43 primary health care teams from the 5 centers participating in the intervention group. Scores in all six areas of the chronic care model improved from the first to the third learning sessions. On a scale of 0–11, the total score increased from 5.7 during LS1 to 8.7 in LS3. The improvements were all statistically significant except in the area of clinical information system. At LS1, the area of decision support received the highest score (7.2) while community linkages obtained the lowest score (4.3). At the end of the intervention during LS3, self-management support received the highest score while community linkages continued to receive the lowest score (6.1), albeit it improved by 1.7 points. The highest increase in the scores was observed in the area of delivery system design (+4.1).

Table 3 presents examples of the qualitative feedback from patients in LS1. In general patients' input indicated that activities for the support group needed to be tailored to different types of patients, and that diabetes education and behavioral changes needed to be reinforced. Overall, the group of patients found the support groups useful to increase physical activity.

Table 4 lists some examples of objectives for improvement activities selected by health care teams to implement the collaborative intervention. These objectives emerged during

Table 3 – Representative examples of contributions from a group of patients participating in LS1.

Views of people with diabetes about the support groups

Strengths:

"The support groups are important because they empower people with diabetes to take leadership roles within their community".

"People can be mutually motivated to carry out some tasks, for example to improve glucose control".

"Support groups are ideal for increasing physical activity".

"The support groups are an aid for the health teams because they complement their care".

"The support groups offer the opportunity to carry out beneficial activities".

Weaknesses:

"Sometimes behavioral changes are not easy to achieve".

"Advice provided in educational activities are not personalized and sometimes are too general".

"The shortage of medicine and the costs of specialized services can not be solved by the support groups".

Table 4 – Examples of objectives[a] for improvement activities by health care teams.

Organization of care

Organize monthly meetings of all core health team members to discuss strategies for patient care, conduct evaluations, provide feedback and establish commitments. The meeting included the participation of patients.

Coordinate nutritional evaluation of all patients.

Implement a patient diabetes educational program.

Strengthen the referral system for all the patients with diabetes.

Monitor an adequate supply of the necessary medication and diabetes testing materials.

Facilitate specialist consultation at the secondary level when needed.

Negotiate with uncontrolled patients their goals to achieve good glycemic control and monitored blood glucose every 15 days.

Establish cross-consultation with specialists in difficult-to-manage cases.

Improvements in insulin use.

Decision support

Carry out referral of patients who do not attain the treatment objectives or are at greater risk.

Follow the protocols established in the Mexican Official Standard of Diabetes.

Train the health care team in the management of diabetic patients.

Prescribe urine exam and lipid profile once a year.

Self-management support

Redesign the support group sessions in a more flexible way with different schedules in order to guarantee at least the monthly attendance of all patients.

Create nucleus of coordinators in education (NCE) (one physician and up to two nurses) in each health center.

Carry out training in self-care for the NCE.

Implement a diabetes education program with adequate educational materials for the NCE and patients.

Implement collective medical consultation for the support groups, promoting the sharing of experiences and problem-solving in diabetes management.

Clinical information system

Reinforce the goals of behavior changes using nurses' notes.

Registry of completeness and readable clinical files.

[a] Selected examples were translated from Spanish. Full list of examples available on request from corresponding author.

Table 5 – Total number of measures (N) and average values[a] in the intervention and usual care groups, of A1c, fasting blood sugar, cholesterol, triglycerides, weight, body mass index, and systolic and diastolic blood pressure, at the beginning and the end of the intervention.

	N	Beginning	End	Difference	p
A1c					
Intervention	196	8.4	7.9	0.5	<0.01
Usual care	111	8.7	8.6	0.1	0.80
Blood glucose					
Intervention	152	163.4	149.1	−14.3	0.01
Usual care	68	178.4	183.1	4.8	0.67
Total cholesterol					
Intervention	195	192.3	177.2	−15.1	<0.01
Usual care	111	200.7	194.4	6.3	0.03
Triglycerides					
Intervention	196	194.4	193.9	−0.4	0.95
Usual care	109	205.4	195.4	−9.9	0.28
Weight					
Intervention	195	65.4	65.4	0.0	1.00
Usual care	110	66.1	68.3	2.1	0.04
BMI					
Intervention	193	28.7	28.7	0.0	0.93
Usual care	107	28.9	29.3	0.3	0.47
SBP					
Intervention	181	120.5	122.1	1.6	0.23
Usual care	99	122.0	122.6	0.6	0.72
DBP					
Intervention	181	74.9	75.2	0.3	0.76
Usual care	99	76.1	78.5	2.4	0.01

[a] Two-tailed paired *t*-test before and after the intervention.

the "Plan" phase of the P-D-S-A cycle. Teams implemented the activities during the "Do" phase (in between learning sessions), then "Studied" the results during the following learning session. The teams then "Acted" on what was learned by making long-term changes in their clinical practice. This allowed teams to test quality improvement activities in the real-work setting and introduce changes in a phased manner. The number of activities implemented by teams by area of the chronic care model varied widely, with the largest number of activities reported for organization of health services and self-management support.

Table 5 shows continuous outcome variables before and after the intervention in both intervention and usual care groups. The initial evaluation indicated that the average A1c was 8.4% and 8.7% in the intervention and usual care groups, respectively. The before–after comparison indicated that the average values of A1c, blood glucose, and cholesterol decreased significantly among the intervention group, but not among the usual care group. The average values of triglycerides, weight, body mass index, and systolic and diastolic blood pressure remained invariable or varied very little in both groups.

Table 6 shows the proportion of patients achieving quality improvement goals before and after the intervention. The proportion of patients with A1c < 7% before the intervention was 27.6% and 20.7% in the intervention and usual care groups, respectively. The comparison of before and after the intervention indicated that the proportion of patients achieving the main goal of the intervention (A1c < 7%) increased significantly among the intervention group compared to the usual care group ($p < 0.05$) as well as those with total cholesterol < 200 mg/dl ($p < 0.01$). The proportion of patients receiving foot and eye exams also showed positive results among the intervention group. Overall the proportion of patients who achieved three or more quality improvement goals increased more than fourfold among the intervention group from 16.6% to 69.7% ($p < 0.01$) while among the usual care group it decreased from 12.4% to 5.9% ($p = 0.12$), although not statistically significant.

9. Discussion

The VIDA experience demonstrates that the chronic care model can be successfully implemented to improve the quality of diabetes care in low-resource settings. In comparison with the usual care group, patients in the intervention group significantly improved their metabolic control, measured by a 0.5% reduction of the glycosylated hemoglobin after 18 months of follow-up. This reduction, albeit modest, is considered clinically significant and comparable to similar trials in developed countries with longer follow-up [13], which also report reduction of fasting blood glucose and cholesterol [14], as does our study. The proportion of patients receiving preventive measures and achieving three or more quality improvement goals was also significantly higher in the intervention group. However, we did not observe differences in blood pressure

Table 6 – Proportion (%) of patients (the intervention group and usual care) meeting quality improvement goals at the beginning and the end of the intervention.

Goal	Beginning		End		p[a]
	N	%	N	%	
A1c < 7% (good control)					
The intervention group	54	27.6	77	39.3	<0.01
Usual care	23	20.7	31	27.9	0.18
p[b]		0.19		0.03	
Cholesterol < 200					
The intervention group	127	65.1	150	76.5	<0.01
Usual care	60	54.1	65	58.6	0.38
p[b]		0.02		0.25	
BP ≤ 140/90					
The intervention group	138	73.4	142	75.1	1.00
Usual care	79	72.5	70	69.3	0.69
p[b]		0.86		0.57	
Foot examination					
The intervention group	96	49.0	187	95.4	<0.01
Usual care	52	46.8	24	21.6	<0.01
p[b]		0.78		<0.01	
Eye examination					
The intervention group	20	10.2	143	73.0	<0.01
Usual care	4	3.6	5	4.5	<0.01
p[b]		0.17		<0.01	
Three or more treatment goals					
The intervention group	33	16.6	137	69.7	<0.01
Usual care	14	12.4	7	5.9	0.12
p[b]		0.68		<0.01	

[a] McNemar Test.
[b] Adjusted for clustering of patients within practice and baseline measures.

control or weight loss, measured by BMI. The combination of better quality care delivered by the health team, greater diabetes self-management, and treatment changes for patients in accordance with Mexican protocols were together responsible for improved outcomes in the intervention group.

An important feature of this intervention was the emphasis on the primary care team. The self-evaluation of the intervention teams revealed improvements in all areas of the chronic care model. The domains of the ACIC survey with higher scores at the end of the intervention were those corresponding to the areas related to major intervention activities such as delivery system design (referral system), self-management support (patient education program) and decision support (clinical in-service training). Consistent with our findings, Kaissi et al. [15] reported that these three domains of the ACIC survey were most frequently correlated with improvements in quality of care measures.

Rather than resort to the traditional referral to hospital, a hospital-based specialist provided support to the team on site. Most importantly, patients became part of the extended team. This was achieved in two ways: first, by improvements in self-management support and delivery system design, which led to strengthen patients' own support groups; and second, by inviting patients to the learning sessions. Specialists' support to primary care has already proven to be effective in the United States [16,17] and the incorporation of patients into the team had been used successfully before by Day in 1992 [18].

9.1. *Limitations of the study*

Although the intervention was highly effective, several weaknesses need to be mentioned to better understand the scope of our findings. First, budget constraints prevented the involvement of more health centers. For the same reason, monitoring patients over a longer period of time was not possible. This prohibited better measurement of the potential benefits that come from sustained glycemic control such as the prevention of the presentation or of the progression of chronic complications. Second, the learning discussion group component of the intervention was optimized by assigning groups based on personality characteristics, rather than by random assignment. Finally, avoiding the "contamination" of centers that acted as controls (those centers providing usual diabetes care) was not possible, because of the visibility and publicity of the intervention at the local level. Despite this limitation, if contamination did occur results would be biased towards the null.

We did not collect data on intermediate process variables, such as adherence to treatment, diet, or physical activity, which would have allowed us to assess what exactly mediated the improvements in the intervention group. However, the multilevel analysis, with the random intercept conditional on the baseline levels of outcome measurements, accounts for the combined effect of all omitted subject-specific covariates that cause some patients to be more prone to improvements than others. In this way, we controlled for possible selection bias.

9.2. *Appropriateness of the study for the setting*

The chronic care model [9,19] and the BTS methodology [20,21] have been applied successfully in numerous interventions targeted at improvements in the care of diabetes and other chronic diseases [15,22–24]. These interventions have primarily taken place in the United States and other developed countries. Developing countries undergoing an epidemiological transition need to adopt models of care more attuned with the health situation of their population. For a long time, external support has over-emphasized infectious disease control based on a model of acute care and vertical interventions. The chronic care model re-conceptualizes primary care to give patients with chronic disease the necessary tools to manage their condition for life.

Although PAHO provided technical support and limited financial resources to implement this intervention, it did not pay any salaries for health providers. The entire process of care was carried out within the envelope of resources available to the health care system of the State of Veracruz.

The project used materials developed in Spanish in Latin America, such as the QUALIDIAB clinical information system and the PENID-LA Diabetes Education program. The PENID-LA Educational Program [11] has been implemented in 10 Latin American countries (Argentina, Bolivia, Brazil, Chile, Colombia, Costa Rica, Cuba, Mexico, Paraguay, and Uruguay). An assessment of the program reported a 1.2% reduction in A1c, but no control group was used [11]. Diabetes education controlled trials in Latin America that used other educational material achieved significant reductions in A1c as well. A previous PAHO/WHO intervention in Chile [25], which included an educational program and behavioral counseling, demonstrated a 0.4% reduction in A1c in the intervention group and a significantly lower 0.1% reduction in the control group after 6 months follow-up. In Costa Rica [26], a study that included nutrition and physical activity interventions managed to decrease A1c by 1.8% in comparison with only 0.4% in the control group. In this study, although participants were randomly assigned to intervention and control groups, the researchers followed stringent selection criteria and implemented a highly intensive intervention.

Reviews by Knight et al. [27] and Shojania et al. [28] reported modest effects of disease management programs on diabetes control, quantified in reduction of A1c, in line with the 0.5% point reported here for VIDA. A study in the United States that used a similar methodology improved process indicators but did not report a reduction of A1c [29]. In Denmark, a randomized study that also directed the intervention to the health system demonstrated a difference in the reduction in A1c at the end of the study of 0.5% between the intervention and control group [30]. Therefore, we consider that the effect we observed to be appropriate for the follow-up period.

There is sufficient evidence of the effectiveness of key interventions for diabetes care [5]; the challenge is on the implementation of these interventions in primary care, using resources that would be readily available in these settings. Through an evaluation process, we have demonstrated it is possible to translate this evidence into practice within a health care infrastructure with limited resources.

Conflict of interest statement

The authors state that they have no conflict of interest.

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